

TWELVE

CONTEXT ENGINEERING

From metrics *to* intelligence.

A MAP OF THE TALK

Three sections, one product story.

01

DEFINING THE LANGUAGE

Model Qualities

The four physical pillars that turn SkillCorner tracking rows into an analytical analytical vocabulary the system — and the user — can reason about. about.

02

THE CONTEXT LAYER

RAG System

How a question becomes an answer: intent classification, retrieval of definitions and player context, prompt assembly, grounded narrative output.

INTENT
RETRIEVE
WORDWISE
REFUSE

03

THE PRODUCT VISION

Storyboard

Three domains, two depths each — the sporting director's loop from a brief, from a brief, into a profile, out to alternatives. A working assistant, not a not a dashboard.

Model Qualities.

Four percentile-based physical dimensions derived from SkillCorner tracking tracking data — transforming raw telemetry into interpretable player profiles. player profiles.

- SPEED
- ACCELERATION
- AGILITY
- ENDURANCE

01

THE FOUR PILLARS

Speed · Acceleration · Agility · Endurance.

Q1

Speed

Peak sprint velocity. Captures both absolute top speed and the ability to sustain it.

PSV-99 (Peak Sprint Velocity)

55%

Top 5 PSV-99

45%

Q2

Acceleration

How quickly a player reaches high speed. Time-to-sprint, time-to-HSR, post-COD acceleration, explosive counts.

Top 3 Time to Sprint

30%

Top 3 Time to HSR

20%

Top 3 Time to Sprint (post-COD)

20%

Top 3 Time to HSR (post-COD)

10%

Explosive Accel to Sprint P90

10%

Explosive Accel to HSR P90

10%

Q3

Agility

Change of direction. Frequency in match plus 505-test times at 90° and 180°.

Change of Direction Count P90

50%

Top 3 Time to 505 @ 90°

25%

Top 3 Time to 505 @ 180°

25%

Q4

Endurance

Sustained output over a match. Metres per minute, total distance, running and high-speed running.

Metres per Minute

30%

Total Distance P90

25%

Running Distance P90

25%

HSR Distance P90

20%

THE FOUR PILLARS

The Physical Scoring Pipeline

Each pillar is computed from aggregated tracking metrics, normalised within position groups, weighted transparently, and converted into percentile-based player profiles.

Aggregate

Per-90 rates, peak metrics and top-N means rolled up across a full season — never a single match.

INPUTS · ~3,200 ROWS / MATCH FLOOR · 450 MIN

01

Normalise

Z-scored within position group. Wide attackers vs wide attackers — never against against centre-backs.

COHORTS · 7 GROUPS METHOD · ROBUST Z

02

Weight

Each pillar is a transparent weighted sum of metric z-scores. No black-box ML, every term is auditable.

FORM · LINEAR SUM / AUDIT · OPEN WEIGHTS

03

Percentile

Output 0–100 percentiles inside the position position group. Built to be quoted in a sentence by a human.

SCALE · 0 — 100 USE · CITE DIRECTLY

04

Retrieval & Context.

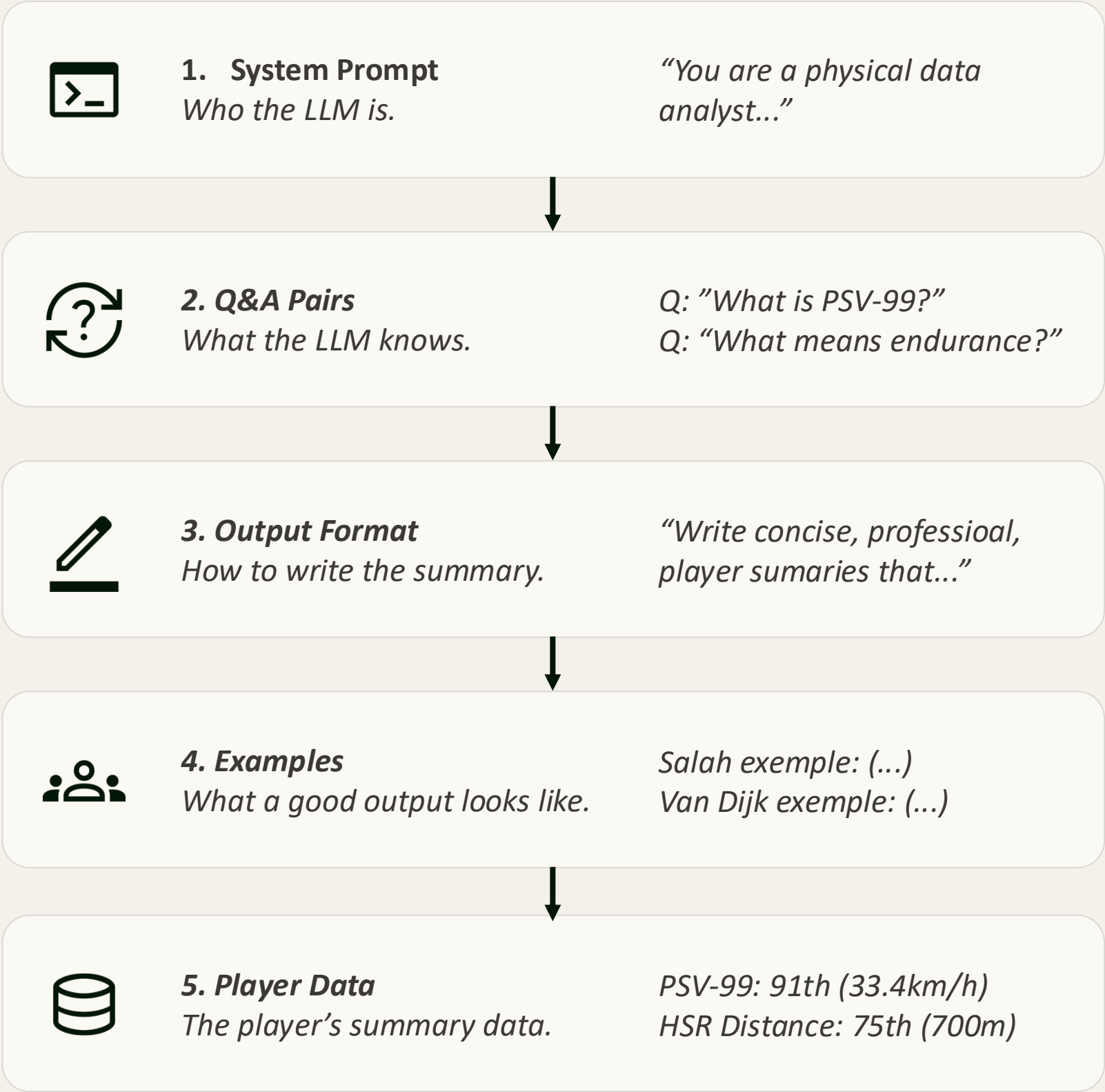
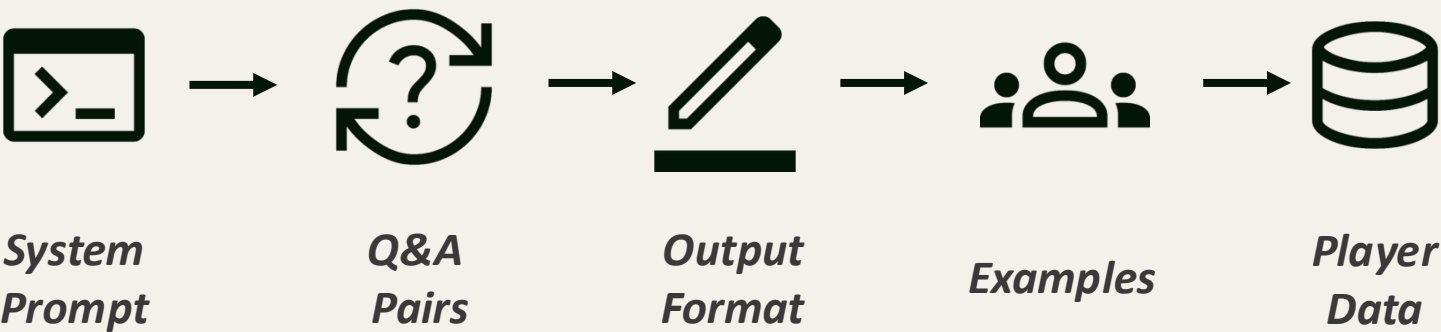
A retrieval-augmented reasoning layer that combines player context, metric definitions, and domain knowledge to generate grounded, explainable football football intelligence.

02

THE FOUR PILLARS

LLM Pipeline

Before the LLM sees the player’s data, it reads a curated knowledge base – a set of Q&Q pairs loaded directly into the conversation history.



54

Q&A PAIRS

Before the LLM sees the player’s data, it reads a curated knowledge base – a set of Q&Q pairs loaded directly into the conversation history.

16 METRICS
DEFINITIONS

What do PSV-99, COD, HSR, Sprint, etc.. actually measure...

X

8 ATTRIBUTE
MEANING

What Speed, Acceleration, Agility, etc... imply.

8 ATTRIBUTE
COMBINATIONS

How pairs of attributes interact (e.g. fast but low endurance)

9 POSITION
CONTEXT

What physical profile suits each position group.

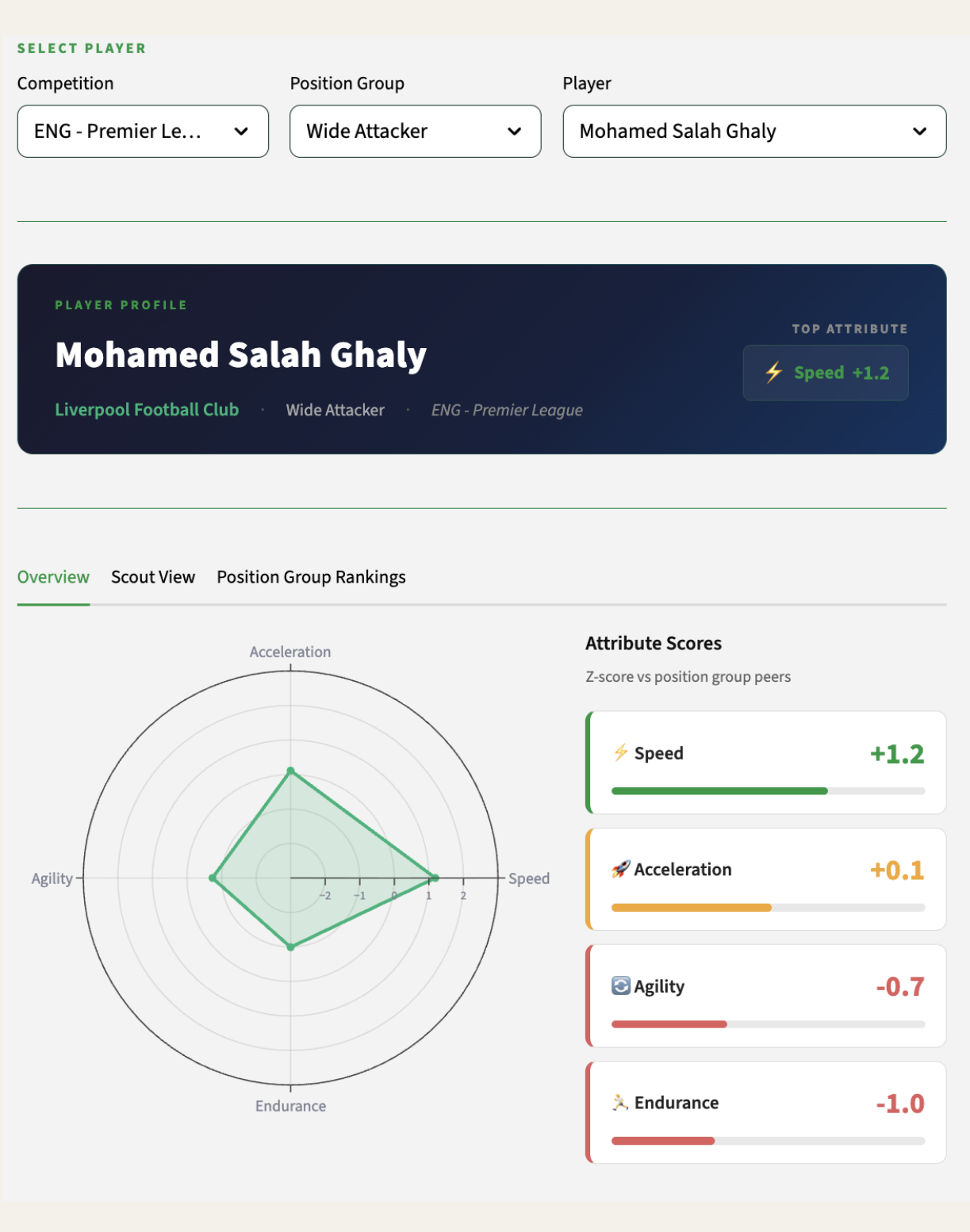
13 INTERPRETATION
GUIDANCE

How to read outliers, inverted metrics, thresholds, etc..

EXAMPLE 01

PLAYER PROFILE

The player profile page displays not only the underlying data — both raw values and percentile-based metrics — but also automatically generated summaries, available in either short or detailed formats.



AI REPORT

Basic Detailed

Regenerate

Physical Summary

Wide attacker with very high top speed and competence in acceleration. Despite fast bursts, he tires sooner and shows below-average distance output, with slower recovery to sprint speed after changing direction.

AI REPORT

Basic Detailed

Regenerate

Physical Summary

In the Premier League this season, Mohamed Salah Ghaly shows a pure speed profile among wide attackers, with very high top speeds and the ability to sustain high speeds in his quickest actions. His acceleration rate is typical for the position, and he is a little stiff through direction changes, indicating some restriction when turning quickly. His stamina is a weak point, with below-average distance metrics and lower metabolic output across matches.

His top speeds are outstanding, and his ability to reach sprint speed quickly remains strong. However, his recovery of sprint speed after a turn is slower than most peers, which can limit rapid re-engagement after a directional change. His metres per minute and total distance are below average, and running distance, along with high-speed running distance, are below the group mean. This profile highlights a player who is blazing fast in bursts but whose sustained output over a full 90 minutes is less pronounced.

EXAMPLE 02

AI PHYSICAL ANALYST

The analyst isn't just a report generator, it's a chat interface. You can ask follow-up questions, compare players, or query the whole dataset. The model routes every question before answering.

12

Hello. Ask about a player, a physical metric, or what is and is not in scope for this analyst.



Analyze Mohamed Salah physical attributes

12

Mohamed Salah Ghaly: As a Wide Attacker, he shows very high top speeds and strong high-speed running capacity, with very high speed maintained on fast efforts. Acceleration is reasonable relative to peers, while agility is poor and direction changes are fewer than most peers. He covers less ground overall and exhibits lower endurance and distance metrics than the typical 187-wide attackers. If you want, I can provide a direct player-to-player comparison or a metric definition for any of these factors.

12

Hello. Ask about a player, a physical metric, or what is and is not in scope for this analyst.



How does Mohamed Salah's speed compare vs Kylian Mbappe?

12

Mohamed Salah Ghaly shows strong top speed relative to his wide-attacker peers, but Mbappé's maximal sprint pace is exceptional for a centre forward. In direct comparison, Mbappé typically records higher peak speeds and sprint efficiency than Salah. If you want, I can give a metric-by-metric breakdown for PSV-99 / TOP-5 PSV-99 figures.

12

Hello. Ask about a player, a physical metric, or what is and is not in scope for this analyst.



Who are the top 5 players by endurance in the dataset?

12

Top 5 fastest (endurance) players in the dataset:

1. Adrien Thomasson Team / role: RC Lens | Midfield Raw Endurance: 100.0

2. Yassine Belkhdim Team / role: Angers SCO | Wide Attacker Raw Endurance: 99.9

3. Daniel Svensson Team / role: Borussia Dortmund | Full Back Raw Endurance: 99.8

4. Matthew Pennington Team / role: Bradford City AFC | Central Defender Raw Endurance: 99.6

5. Lovro Majer Team / role: VfL Wolfsburg | Midfield Raw Endurance: 99.5

Note: Endurance is the dataset's peak-velocity proxy.

Future Vision Storyboard.

03

The receptionist routes to six functions

DOMAIN 1 · PROFILE

Who is this player physically

SUMMARY

getPhysicalProfile(player)

Deterministic. Fixed shape.

ASK

askAboutProfile(player, q)

Intent-driven. Receptionist filters off-scope upstream.

DOMAIN 2 · SIMILARITY / COMPARE

Player ↔ others

SUMMARY

findSimilarOrCompare(mode, A, B?)

Deterministic. Fixed shape.

ASK

askAboutSimilarity(q)

Intent-driven. Receptionist filters off-scope upstream.

DOMAIN 3 · SHORTLIST

Brief → candidates

SUMMARY

shortlistByAttribute(metric, pos, league, n)

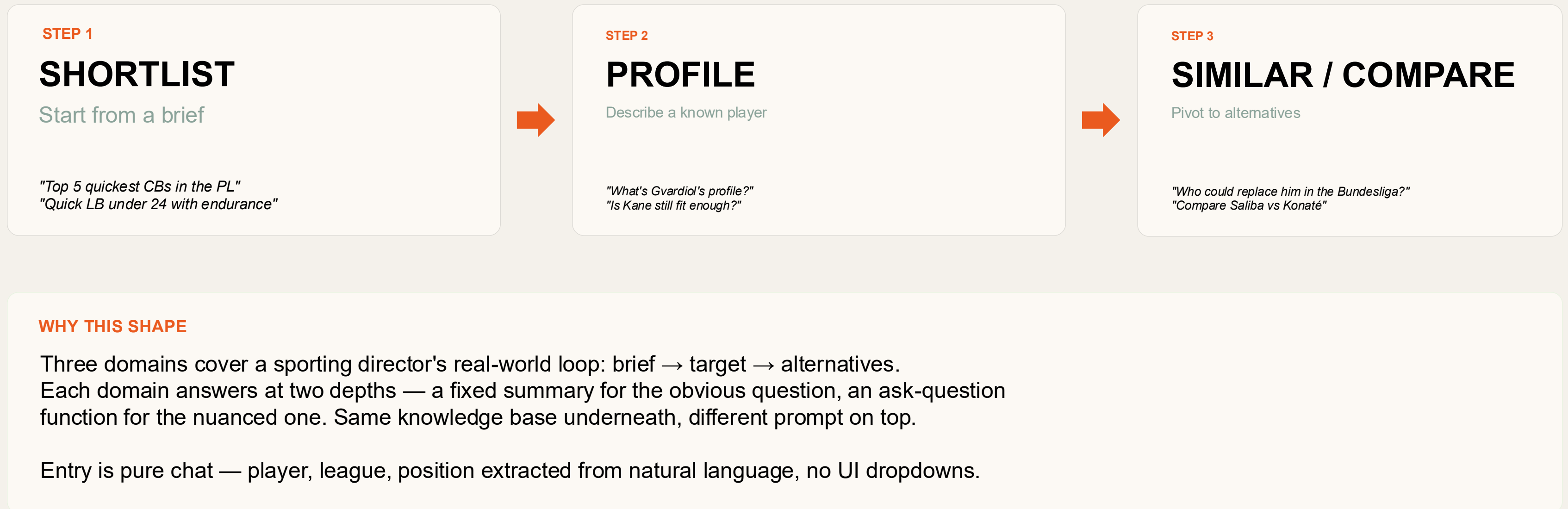
Deterministic. Fixed shape.

ASK

askShortlistQuestion(brief)

Intent-driven. Receptionist filters off-scope upstream.

One loop, three entry points



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CONTEXT ENGINEERING

ENGINEERING

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